# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2002-I-Q02] | Part A (Q1-Q5)**

Twenty equal circles fill a rectangle in three staggered rows of $7, 6, 7$ circles. Each outer row is tangent to both vertical sides, each row has neighboring circles tangent, and each middle-row circle touches the two nearest circles in each outer row. The top and bottom circles touch the horizontal sides. If the rectangle aspect ratio is $(\sqrt{p} - q)/2$ with positive integers $p, q$, find $p + q$.

二十个等圆在矩形内排成三行，数目为 $7, 6, 7$，中间一行交错放置。外两行各与左右边相切；同行邻圆相切，中行每圆与上下行各两个最近圆相切；最上、最下圆分别切水平边。若长宽比为 $(\sqrt{p} - q)/2$，求正整数 $p, q$ 之和。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2005-I-Q01] | Part A (Q1-Q5)**

Six equal circles form a cyclic ring, each touching its neighbors externally and a radius-$30$ circle internally. If $K$ is the area inside the large circle but outside the six small disks, find $\lfloor K \rfloor$.

六个等圆首尾成环，相邻外切，且均与半径 $30$ 的大圆内切。大圆内、六小圆盘外的面积为 $K$，求 $\lfloor K \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2009-II-Q05] | Part A (Q1-Q5)**

An equilateral triangle is inscribed in a radius-ten circle. At its three vertices place internally tangent circles with radii $3, 2, 2$, respectively. A fourth circle is externally tangent to all three. If its radius is reduced $m/n$, find $m + n$.

半径十的圆内接等边三角形，在三个顶点分别作与大圆内切的半径三、二、二小圆。另一圆与三个小圆均外切，半径为最简分数 $m/n$，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2000-II-Q10] | Part B (Q6-Q10)**

A circle is inscribed in quadrilateral $ABCD$, touching $AB$ at $P$ and $CD$ at $Q$. If $AP = 19, PB = 26, CQ = 37, QD = 23$, find the circle radius squared.

圆内切于四边形 $ABCD$，与 $AB, CD$ 分别切于 $P, Q$。已知 $AP = 19, PB = 26, CQ = 37, QD = 23$，求半径平方。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2005-II-Q08] | Part B (Q6-Q10)**

Two externally tangent circles of radii $4, 10$ are internally tangent to a third circle, all centers collinear. A common external tangent cuts a chord of the third circle of length $m\sqrt{n}/p$ in reduced squarefree form. Find $m+n+p$.

半径 $4, 10$ 的两圆外切，并均与第三圆内切，三圆心共线。两小圆的公外切线截大圆的弦长为约简的 $m\sqrt{n}/p$，求 $m+n+p$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2006-II-Q09] | Part B (Q6-Q10)**

Circles centered at $(0,0),(12,0),(24,0)$ have radii $1,2,4$. Choose a positive-slope common internal tangent of the first two, and a negative-slope common internal tangent of the last two. Their intersection has $x = p - q\sqrt{r}$ in squarefree form. Find $p+q+r$.

圆心为 $(0,0),(12,0),(24,0)$ 的三圆半径分别为一、二、四。取前两圆正斜率公内切线及后两圆负斜率公内切线，交点横坐标为 $p - q\sqrt{r}$，求 $p+q+r$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2007-I-Q09] | Part B (Q6-Q10)**

A right triangle has legs $30, 16$. On the hypotenuse's exterior side, two equal circles touch one another externally; one is tangent to the hypotenuse and the extension of the first leg, the other to the hypotenuse and the extension of the second leg. If their radius is reduced $p/q$, find $p+q$.

直角三角形两直角边长为三十、十六。在斜边外侧作两个等圆相互外切，并分别与斜边及两条直角边的向外延长线相切。半径为最简分数 $p/q$，求 $p+q$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2014-II-Q08] | Part B (Q6-Q10)**

A radius-two circle has diameter $AB$. An inner circle $D$ is tangent to it at $A$. A circle $E$ is internally tangent to the outer circle, externally tangent to $D$, and tangent to segment $AB$. The radius of $D$ is three times that of $E$ and equals $\sqrt{m}-n$. Find $m+n$.

半径二的圆以 $AB$ 为直径，内圆 $D$ 与它在 $A$ 内切。圆 $E$ 与大圆内切、与 $D$ 外切，并与直径线段相切。$D$ 半径为 $E$ 的三倍，且为 $\sqrt{m}-n$，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2022-I-Q08] | Part B (Q6-Q10)**

An equilateral triangle $ABC$ has circumradius $18$. For each vertex, draw the circle tangent to both adjacent side segments and internally tangent to the circumcircle. The three circles meet pairwise at six points. The three intersections farther from the vertices form the smaller of two concentric equilateral triangles. Its side is $\sqrt{a}-\sqrt{b}$ for positive integers $a, b$. Find $a+b$.

等边三角形 $ABC$ 的外接圆半径为 $18$。在每个顶点处作一个圆，与两条邻边线段相切并与外接圆内切。这三个圆两两交于六点；其中离三个原顶点较远的三个交点构成较小的等边三角形。其边长为 $\sqrt{a}-\sqrt{b}$，$a, b$ 为正整数。求 $a+b$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2022-II-Q07] | Part B (Q6-Q10)**

Two circles of radii $6$ and $24$ touch externally. Their three common tangent lines bound a triangle. Find its area.

半径为 $6, 24$ 的两圆外切，三条公切线围成一个三角形。求其面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2024-I-Q08] | Part B (Q6-Q10)**

Inside triangle $ABC$, a row of eight mutually consecutive tangent circles of radius $34$ rests on $BC$, with the end circles touching $AB$ and $AC$. A row of $2024$ circles of radius $1$ fits in the same way. If the triangle inradius is $m/n$ in lowest terms, find $m+n$.

在三角形 $ABC$ 内，八个半径 $34$ 的圆依次相切且均与 $BC$ 相切，首尾圆分别与 $AB, AC$ 相切。同样方式也能排下 $2024$ 个半径 $1$ 的圆。内切圆半径为最简分数 $m/n$，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2025-II-Q06] | Part B (Q6-Q10)**

A radius-$6$ circle centered at $A$ is internally tangent at $B$ to a radius-$15$ circle. Let $BC$ be a large-circle diameter, and $D$ a large-circle point with $AD \perp BC$. A rectangle $EFGH$ is inscribed in the small circle, with $EF \perp BC$, $C$ nearer $GH$ than $EF$, and $D$ nearer $FG$ than $EH$. If triangles $DGF, CHG$ have equal area, find $m+n$ for rectangle area $m/n$ in lowest terms.

半径 $6$、圆心为 $A$ 的圆与半径 $15$ 的圆在 $B$ 内切。$BC$ 为大圆直径，$D$ 在大圆上且 $AD \perp BC$。小圆内接矩形 $EFGH$ 满足 $EF \perp BC$，$C$ 更靠近 $GH$，$D$ 更靠近 $FG$。若 $DGF, CHG$ 面积相等，将矩形面积化为最简分数 $m/n$，求 $m+n$。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2002-II-Q14] | Part C (Q11-Q15)**

Right triangle $APM$, right at $A$, has perimeter $152$. A circle of radius $19$ has center $O \in AP$ and is tangent to $AM, PM$. If $OP = m/n$ reduced, find $m + n$.

直角三角形 $APM$ 的直角在 $A$，周长为 $152$。半径 $19$ 的圆心 $O$ 位于 $AP$ 上，并与 $AM, PM$ 相切。若 $OP = m/n$ 已约分，求 $m + n$。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2002-II-Q15] | Part C (Q11-Q15)**

Two circles intersect at two points, one being $(9, 6)$, and their radii have product $68$. Both the $x$-axis and $y = mx$ with $m > 0$ are common tangents. Write $m = a\sqrt{b}/c$ with $b$ squarefree and $\gcd(a, c) = 1$. Find $a + b + c$.

两圆交于两点，其中一点为 $(9, 6)$，半径乘积为 $68$。$x$ 轴及 $y = mx$（$m > 0$）均为两圆公切线。写 $m = a\sqrt{b}/c$，$b$ 无平方因子且 $a, c$ 互质，求参数和。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2004-II-Q12] | Part C (Q11-Q15)**

An isosceles trapezoid has bases $AB = 6, CD = 4$ and legs $5$. Circles centered at $A, B$ have radius $3$; those at $C, D$ have radius $2$. A circle inside the trapezoid touches all four. If its radius is $(-k + m\sqrt{n})/p$ with squarefree $n$ and $\gcd(k, p) = 1$, find $k + m + n + p$.

等腰梯形两底为 $AB = 6, CD = 4$，腰均为五。以 $A, B$ 为心的圆半径三，以 $C, D$ 为心的圆半径二。梯形内一圆与四圆都相切。半径写为 $(-k + m\sqrt{n})/p$，$n$ 无平方因子且 $k, p$ 互质，求参数和。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2007-II-Q15] | Part C (Q11-Q15)**

Inside a $13$–$14$–$15$ triangle, three equal circles each touch the two sides at one vertex, and a fourth equal circle is externally tangent to all three. If their radius is reduced $m/n$, find $m+n$.

在边长十三、十四、十五的三角形内，三个等圆各与某顶点的两条邻边相切，第四个等圆分别与三者外切。若公共半径为最简分数 $m/n$，求 $m+n$。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2008-II-Q11] | Part C (Q11-Q15)**

In triangle $ABC$, $AB = AC = 100, BC = 56$. A radius-sixteen circle touches $AC, BC$. A second circle touches $AB, BC$, lies wholly inside the triangle, and is externally tangent to the first. If its radius is $m - n\sqrt{k}$ with squarefree $k$, find $m + nk$.

等腰三角形 $ABC$ 中 $AB = AC = 100, BC = 56$。半径十六的圆与 $AC, BC$ 相切，另一圆与 $AB, BC$ 相切且完全在三角形内，并与前圆外切。其半径为 $m - n\sqrt{k}$，求 $m + nk$。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2015-II-Q15] | Part C (Q11-Q15)**

Circles of radii $1$ and $4$ touch externally at $A$. A common external tangent touches them at $B, C$, respectively. A line through $A$ meets the small circle again at $D$ and the large one at $E$, with $B, C$ on the same side of this line. Suppose $[DBA] = [ACE] = m/n$ in lowest terms. Find $m + n$.

半径为 $1, 4$ 的两圆外切于 $A$，一条外公切线分别切两圆于 $B, C$。过 $A$ 的直线再次交小圆、大圆于 $D, E$，且 $B, C$ 在该直线同侧。若 $[DBA] = [ACE] = m/n$ 为最简分数，求 $m + n$。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2019-I-Q11] | Part C (Q11-Q15)**

An integer-sided isosceles triangle has $AB = AC$. A circle centered at its incenter is internally tangent to the $BC$-excircle and externally tangent to both other excircles. Find the least possible triangle perimeter.

整边等腰三角形 $AB = AC$，某以其内心为圆心的圆与 $BC$ 旁切圆内切、与另两旁切圆外切。求最小周长。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2020-II-Q13] | Part C (Q11-Q15)**

A convex pentagon has consecutive side lengths $5, 6, 6, 6, 7$ and a circle tangent to all five sides. Determine its area.

凸五边形的连续五边长为 $5, 6, 6, 6, 7$，且存在与五边均相切的内切圆。求面积。

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# Section 2.6 Tangent circles & common tangents

相切圆与公切线

**[AIME-2021-I-Q13] | Part C (Q11-Q15)**

Two intersecting circles have radii $961, 625$ and common chord line $AB$. Another circle is externally tangent to both, and line $AB$ cuts it in a chord whose minor arc measures $120^\circ$. Find the distance between the first two centers.

半径为 $961, 625$ 的两圆相交于 $A, B$。第三圆与前两圆均外切，直线 $AB$ 截第三圆所得弦的劣弧为 $120^\circ$。求前两圆圆心距。

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